

◦ 7.1 Integration by parts

$$\circ \int f(x) g'(x) dx = f(x) g(x) - \int g(x) f'(x) dx$$

or

$$\circ \int u dv = uv - \int v du$$

◦ Definite integrals:

$$\int_a^b f(x) g'(x) dx = [f(x) g(x)]_a^b - \int_a^b g(x) f'(x) dx$$

$$\text{or } \int_a^b u dv = [uv]_a^b - \int_a^b v du$$

◦ 7.2 Trigonometric integrals

◦ For $\int \sin^m x \cos^n x dx$

↳ If n is odd, i.e. $n = 2k+1$

$$\int \sin^m x \cos^{2k+1} x dx = \int \sin^m x (1 - \sin^2 x)^k \cos x dx$$

↳ If m is odd, i.e. $m = 2k+1$

$$\int \sin^{2k+1} x \cos^n x dx = \int \cos^n x (1 - \cos^2 x)^k \sin x dx$$

↳ If both m, n are even use $\frac{1}{2}$ angle formulas.

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x), \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x), \quad \sin 2x \cos x = \frac{1}{2} \sin 2x$$

◦ For $\int \tan^m x \sec^n x dx$

↳ If n is even, i.e. $n = 2k, k \geq 2$

$$\int \tan^m x \sec^{2k} x dx = \int \tan^m x (1 + \tan^2 x)^{k-1} \sec^2 x dx \quad \ln |\csc \theta - \cot \theta|$$

↳ If m is odd, i.e. $m = 2k+1$,

$$\int \tan^{2k+1} x \sec^n x dx = \int (\sec^2 x - 1)^k \sec^{n-1} x \tan x \sec x dx$$

◦ Misc.

$$\int \tan x dx = \ln |\sec x| + C = -\ln |\cos x| + C \quad \int \sec x dx = \ln |\tan x + \sec x| + C$$

$$\int \cot x dx = -\ln |\csc x| + C = \ln |\sin x| + C \quad \int \csc x dx = -\ln |\csc x + \cot x| + C$$

Product Identities

$$\sin A \cos B = \frac{1}{2} [\sin(A-B) + \sin(A+B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

7.3 Trigonometric substitution

For $\sqrt{a^2 - x^2}$, sub $x = a \sin \theta \Rightarrow \sin \theta = \frac{x}{a}$, $\cos \theta = \frac{1}{a} \sqrt{a^2 - x^2}$, $dx = a \cos \theta$

For $\sqrt{a^2 + x^2}$, sub $x = a \tan \theta \Rightarrow \tan \theta = \frac{x}{a}$, $\sec \theta = \frac{1}{a} \sqrt{a^2 + x^2}$, $dx = a \sec^2 \theta$

For $\sqrt{x^2 - a^2}$, sub $x = a \sec \theta \Rightarrow \sec \theta = \frac{x}{a}$, $\tan \theta = \frac{1}{a} \sqrt{x^2 - a^2}$, $dx = a \tan \theta \sec \theta$

7.4 Partial fractions

For $f(x) = \frac{P(x)}{Q(x)}$

↳ If $\deg(P) \geq \deg(Q)$, f is improper

• $f(x) = S(x) + \frac{R(x)}{Q(x)}$ by long division
 Polynomial proper

↳ If $\deg(P) < \deg(Q)$, f is proper

• 1.) $Q(x) = (a_1x + b_1)(a_2x + b_2) \dots (a_kx + b_k)$

$$\frac{R(x)}{Q(x)} = \frac{A_1}{(a_1x + b_1)} + \frac{A_2}{(a_2x + b_2)} + \dots + \frac{A_k}{(a_kx + b_k)}, A_n \text{ are constants}$$

• 2.) $Q(x) = (a_1x + b_1)^n, n \in \mathbb{Z}^+$

$$\frac{R(x)}{Q(x)} = \frac{A_1}{(a_1x + b_1)} + \frac{A_2}{(a_1x + b_1)^2} + \dots + \frac{A_k}{(a_1x + b_1)^k}, A_n \text{ are constants}$$

• 3.) $Q(x) = (ax^2 + bx + c)(dx^2 + ex + f) \dots$, $ax^2 + bx + c$ is irreducible

$$\frac{R(x)}{Q(x)} = \frac{Ax + B}{ax^2 + bx + c} + \frac{Cx + D}{dx^2 + ex + f} + \dots$$

• 4.) $Q(x) = (ax^2 + bx + c)^n, n \in \mathbb{Z}^+$

$$\frac{R(x)}{Q(x)} = \frac{Ax + B}{ax^2 + bx + c} + \frac{Cx + D}{(ax^2 + bx + c)^2} + \dots$$

(Probably optimal) Weierstrass Substitution for rational functions w/ $\sin x$, $\cos x$.

$$\text{sub } t = \tan(x/2)$$

$$\hookrightarrow \cos \frac{x}{2} = \frac{1}{\sqrt{1+t^2}}, \quad \sin \frac{x}{2} = \frac{t}{\sqrt{1+t^2}}$$

$$\cos x = \frac{1-t^2}{1+t^2}, \quad \sin x = \frac{2t}{1+t^2}$$

$$dx = \frac{2}{1+t^2} dt$$

Complex numbers.

$$x^2 + 1 = (x+i)(x-i)$$

and

$$\ln(a+ib) = \ln \sqrt{a^2+b^2} + i \arctan\left(\frac{b}{a}\right)$$

7.5 Guidelines of integration

• Inverse trig integrals:

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right), \text{ for } a > 0$$

• Other integrals:

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

$$\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln |x + \sqrt{x^2 \pm a^2}|$$

7.8 Improper Integrals

• Infinite integrals:

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$$

$\hookrightarrow$ If limit exists, the integral is convergent
if limit DNE, the integral is divergent

If both integrals are convergent,

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^\infty f(x) dx, \quad a \text{ is any real number.}$$

◦ Discontinuous integrals:

If f is cont on $[a, b)$,

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

If f is cont on $(a, b]$,

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$$

If both limit exists, (integrals are convergent),

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

where f is dis cont. at some c , $a < c < b$

Comparison Theorem:

Suppose f and g are cont functions, $f(x) \geq g(x) \geq 0$ for $x \geq a$

↳ If $\int_a^\infty f(x) dx$ converges $\longrightarrow \int_a^\infty g(x) dx$ converges

↳ If $\int_a^\infty g(x) dx$ diverges $\longrightarrow \int_a^\infty f(x) dx$ diverges

◦ 8.1 Arc Length

◦ Arc length:

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx \quad \text{or} \quad L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

◦ 8.2 Surface area

◦ Surface area about x -axis:

$$S = \int_a^b \underbrace{2\pi f(x)}_{\text{circumference}} \sqrt{1 + [f'(x)]^2} dx$$

◦ Surface area about y -axis:

$$S = \int_c^d 2\pi g(y) \sqrt{1 + [g'(y)]^2} dy$$

8.3 Applications to Physics and Engineering

Hydrostatic Pressure and Force

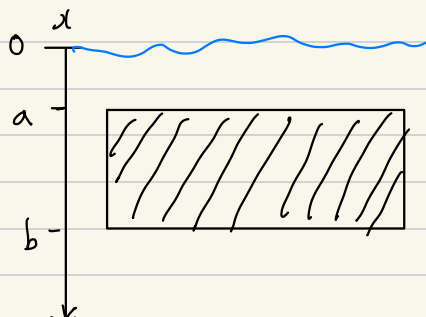
↳ Pressure = (Density) (Gravitational const.) (depth)

$$P = \rho g d$$

↳ Force = (Pressure) (Area)

$$= \int P \, dA$$

$$= \int_a^b \rho g x w(x) \, dx$$



Center of Mass:

↳ Additivity principle:

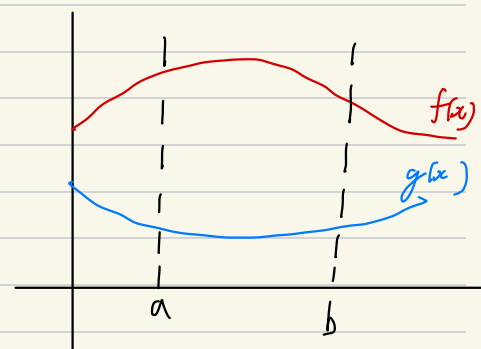
$$\bar{x} = \bar{x}_1 \frac{A_1}{A} + \bar{x}_2 \frac{A_2}{A} + \bar{x}_3 \frac{A_3}{A} \dots$$

$$\bar{y} = \bar{y}_1 \frac{A_1}{A} + \bar{y}_2 \frac{A_2}{A} + \bar{y}_3 \frac{A_3}{A} \dots, \quad A = \sum_{i=1}^n A_i$$

↳ Formula for CM:

$$\bar{x} = \frac{1}{A} \int_a^b x f(x) \, dx$$

$$\bar{y} = \frac{1}{A} \int_a^b \frac{1}{2} [f(x)]^2 \, dx$$



↳ Formula for CM b/w 2 curves

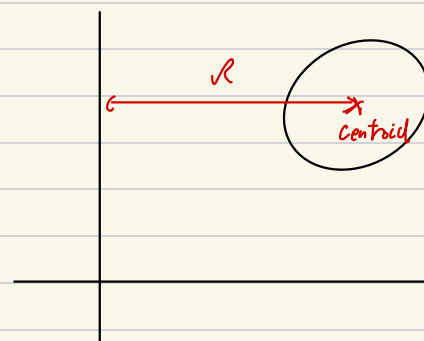
$$\bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] \, dx, \quad A = \int_a^b [f(x) - g(x)] \, dx$$

$$\bar{y} = \frac{1}{A} \int_a^b \frac{1}{2} [f(x)^2 - g(x)^2] \, dx$$

Theorem of Pappus

$$V = 2\pi \bar{R} A, \quad \bar{R} = \text{distance from centroid of shape}$$

↳ volume of revolution



6.7 Hyperbolic Functions

Definitions:

$$\hookrightarrow \sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\hookrightarrow \tanh x = \frac{\sinh x}{\cosh x}$$

$$\hookrightarrow \operatorname{csch} x = \frac{1}{\sinh x} \quad \text{and} \quad \text{etc.}$$

Properties:

$$\hookrightarrow \sinh(-x) = -\sinh x$$

$$\hookrightarrow \cosh(-x) = \cosh x$$

$$\hookrightarrow \cosh^2 x - \sinh^2 x = 1$$

$$\hookrightarrow 1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\hookrightarrow \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$$

$$\hookrightarrow \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

Derivatives:

$$\hookrightarrow (\sinh x)' = \cosh x$$

$$\hookrightarrow (\cosh x)' = \sinh x$$

$$\hookrightarrow (\tanh x)' = \operatorname{sech}^2 x$$

$$\hookrightarrow (\operatorname{csch} x)' = -(\operatorname{csch} x)(\coth x)$$

$$\hookrightarrow (\operatorname{sech} x)' = -(\operatorname{sech} x)(\tanh x)$$

$$\hookrightarrow (\coth x)' = -\operatorname{csch}^2 x$$

Inverse:

$$\hookrightarrow \sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

$$\hookrightarrow \cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$$

$$\hookrightarrow \tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$$

Inverse Derivatives:

$$\hookrightarrow (\sinh^{-1} x)' = \frac{1}{\sqrt{1+x^2}}$$

$$\hookrightarrow (\cosh^{-1} x)' = \frac{1}{\sqrt{x^2 - 1}}$$

$$\hookrightarrow (\tanh^{-1} x)' = \frac{1}{1-x^2}$$

$$\hookrightarrow (\operatorname{csch}^{-1} x)' = \frac{1}{|x|\sqrt{x^2 + 1}}$$

$$\hookrightarrow (\operatorname{sech}^{-1} x)' = -\frac{1}{x\sqrt{1-x^2}}$$

$$\hookrightarrow (\coth^{-1} x)' = \frac{1}{1-x^2}$$

10.1 Curves Defined by Parametric Equations

Definition:

t is a parameter s.t. $x = f(t)$, $y = g(t)$
If $a \leq t \leq b$, the initial point is $[f(a), g(a)]$
the terminal point is $[f(b), g(b)]$

Circle, Eclipse:

Circle: $x = r \cos t$, $y = r \sin t$ ($x^2 + y^2 = r^2$)
Eclipse: $x = a \cos t$, $y = b \sin t$ ($\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$)

10.2 Calculus with Parametric Curves

Tangents

$$\hookrightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \quad \frac{dx}{dt} \neq 0$$

- Vertical when $dx/dt = 0$

- Horizontal when $dy/dt = 0$

- Apply L' Hospital Rule when $\frac{dx}{dt} = \frac{dy}{dt} = 0$

$$\hookrightarrow \frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

Area

Area along x axis:

$$\hookrightarrow A_x = \int_{t_1}^{t_2} y(t) x'(t) dt$$

Area within a curve:

$$\left| \int_{t_0}^{t_1} x(t) y'(t) dt \right| \quad \text{or} \quad \left| \int_{t_0}^{t_1} y(t) x'(t) dt \right|$$

where t_0, t_1 have equal x, y coordinates.

Area along y axis:

$$\hookrightarrow A_y = \int_{t_1}^{t_2} x(t) y'(t) dt$$

Arc length

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt$$

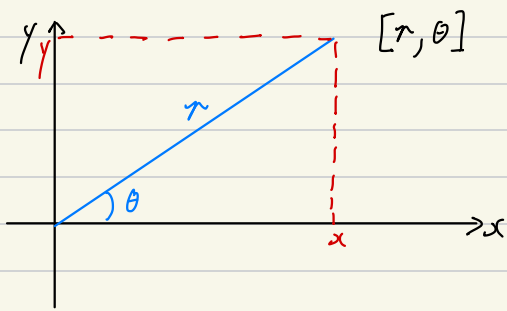
Surface area

$$S = \int_{t_1}^{t_2} 2\pi y(t) \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt \quad \text{about } x \text{ axis}$$

$$S = \int_{t_1}^{t_2} 2\pi x(t) \sqrt{\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2} dt \quad \text{about } y \text{ axis}$$

10.3 Polar coordinates

Polar coordinates:



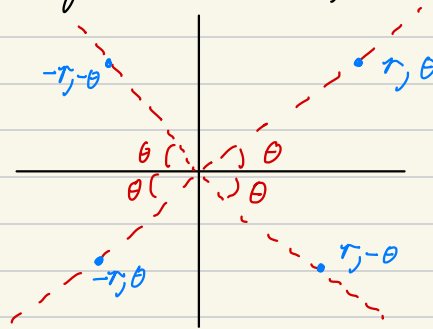
$$r = \pm \sqrt{x^2 + y^2}$$

$$y = r \sin \theta$$

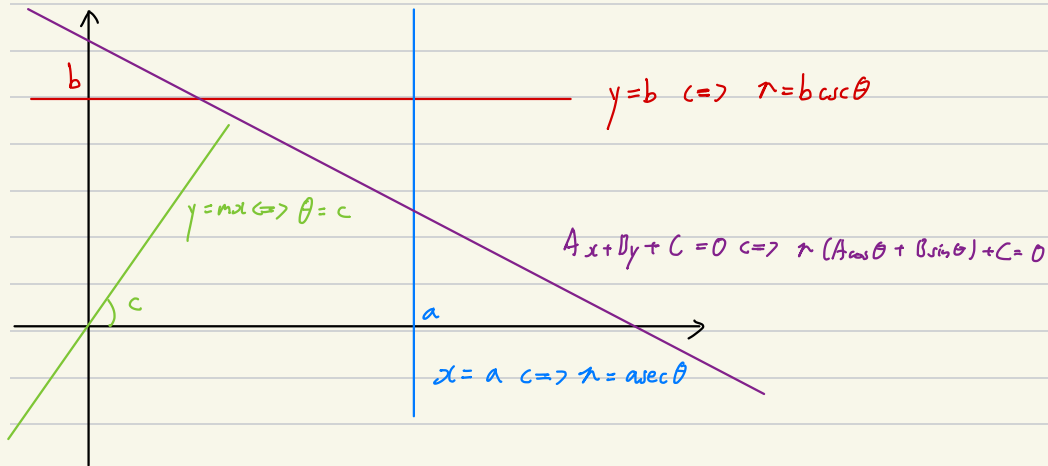
$$x = r \cos \theta$$

$$\theta = \arctan \frac{y}{x}$$

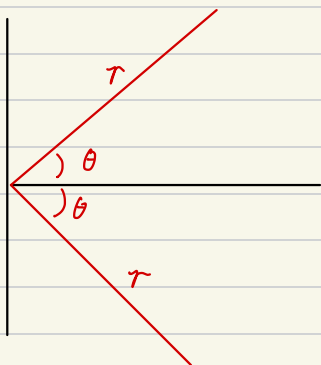
Negative and r, θ



Basic graphs:

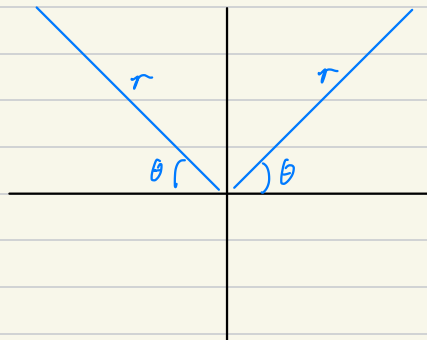


Symmetry



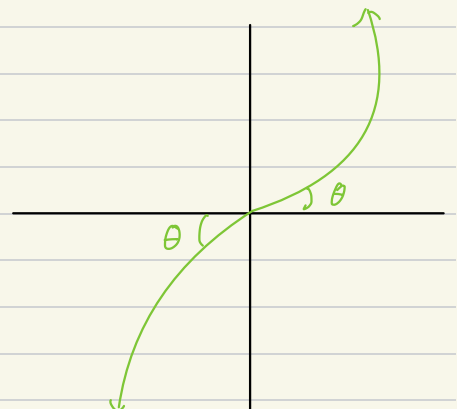
Symmetry by x-axis

$$[r, \theta] = [r, -\theta]$$



Symmetry by y-axis

$$[r, \theta] = [r, \pi - \theta]$$



Symmetry by origin

$$[r, \theta] = [r, \pi + \theta]$$

Graphing in polar coordinates

1.) Find θ when $r=0$

2.) Find $\frac{dr}{d\theta}$,

set $\frac{dr}{d\theta} = 0$, $\frac{dr}{d\theta} = DNE$

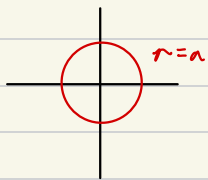
3.) Find symmetry

4.) Note interesting points $(0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \pi)$

Common Polar Curves

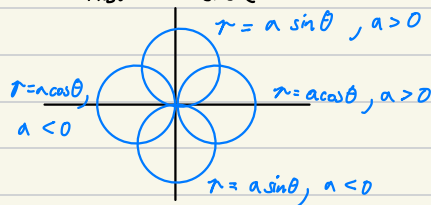
Circle, spiral

Circle

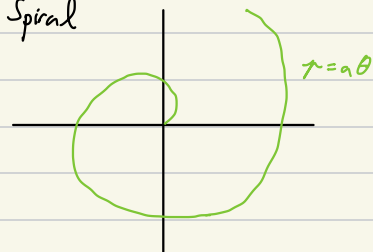


- Also

circle



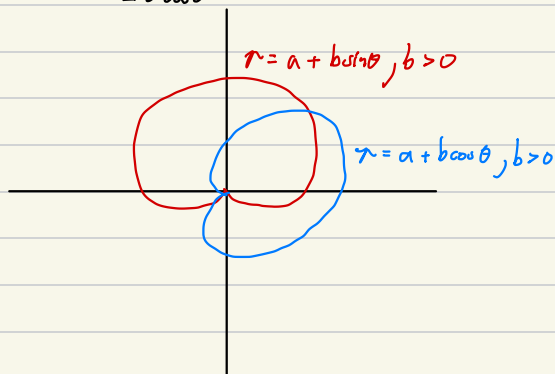
- Spiral



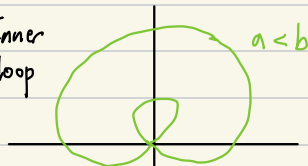
Limacons

$$r = a \pm b \sin \theta$$

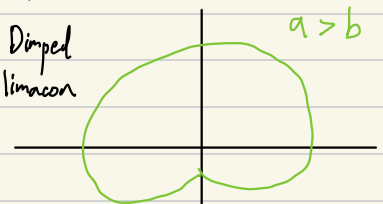
$$r = a \pm b \cos \theta$$



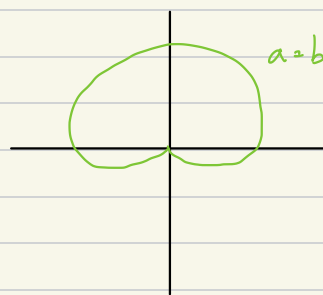
- Inner loop



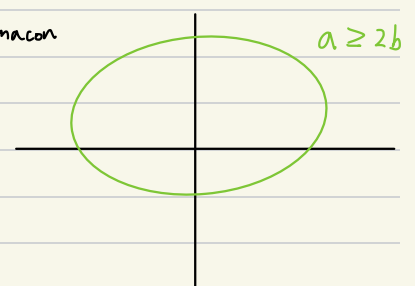
- Dimpled limacon



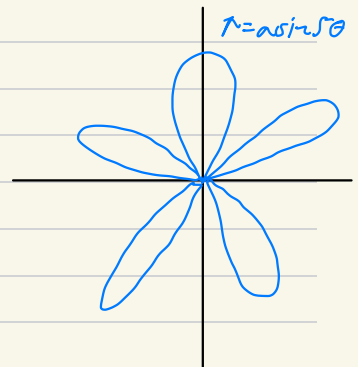
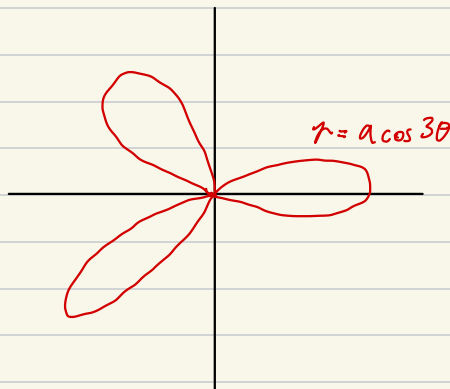
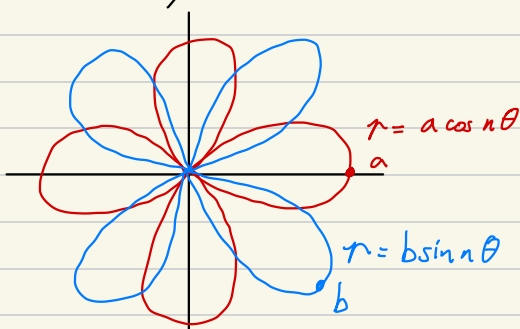
- Cardioid



- Convex limacon

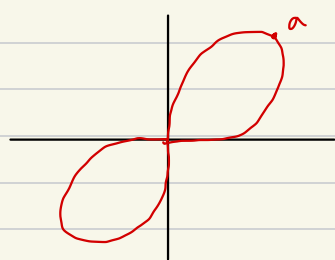


Petal (Roses), $n \in \mathbb{Z}^+$

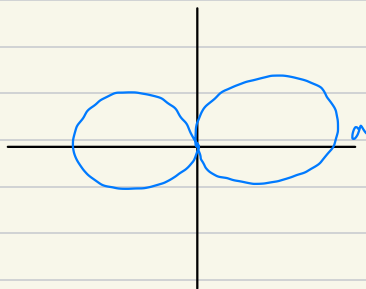


$\hookrightarrow n$ odd, n petals
 $\hookrightarrow n$ even, $2n$ petals

◦ Lemniscates



$$r^2 = a^2 \sin 2\theta$$



$$r^2 = a^2 \cos 2\theta$$

◦ 10.4 Calculus in Polar Coordinates

◦ Area:

◦ Area b/w 2 curves

$$A = \int_a^b \frac{1}{2} r^2 d\theta$$

$$A = \int_a^b \frac{1}{2} ([f(\theta)]^2 - [g(\theta)]^2) d\theta$$

◦ Arc Length

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

◦ Tangents

$$\hookrightarrow \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

Horizontal when $\frac{dy}{d\theta} = 0$
Vertical when $\frac{dx}{d\theta} = 0$

At $r=0$:

$$\hookrightarrow \frac{dy}{dx} = \tan \theta$$

$\hookrightarrow$ If $\frac{dy}{d\theta} = \frac{dx}{d\theta} = 0$:

Use L' Hospital.

◦ 11.1 Sequences

◦ Increasing / Decreasing

$\hookrightarrow a_n$ increasing iff $a_n < a_{n+1}$

$\hookrightarrow a_n$ non-decreasing iff $a_n \leq a_{n+1}$

$\hookrightarrow a_n$ decreasing iff $a_n > a_{n+1}$

$\hookrightarrow a_n$ non-increasing iff $a_n \geq a_{n+1}$

monotonic sequences

◦ Bounded above / Bounded below

$\hookrightarrow$ Bounded above if:

A number M exists such that $a_n \leq M$ for all $n \geq 1$

$\hookrightarrow$ Bounded below if:

A number m exists such that $a_n \geq m$ for all $n \geq 1$

$\hookrightarrow$ Bounded Sequence if bounded above and below.

◦ Limit of sequence

Definition: $\lim_{n \rightarrow \infty} a_n = L$

$\hookrightarrow a_n$ converges if L exists
diverges if L DNE

Rigorous Definition:

$\hookrightarrow \lim_{n \rightarrow \infty} a_n = L$ if
for every $\epsilon > 0$, an integer N exists such that
if $n > N$, $|a_n - L| < \epsilon$

$\hookrightarrow \lim_{n \rightarrow \infty} a_n = \infty$, if

for every positive M there exists an integer N s.t.
if $n > N$ then $a_n > M$

◦ Theorems:

$\hookrightarrow$ If $\lim_{x \rightarrow \infty} f(x) = L$ and $f(n) = a_n$ when n is an integer,
then $\lim_{n \rightarrow \infty} a_n = L$

$\hookrightarrow$ Limit Laws for Sequences, let $\lim_{n \rightarrow \infty} a_n = L$, $\lim_{n \rightarrow \infty} b_n = G$

$\hookrightarrow \lim_{n \rightarrow \infty} (a_n + b_n) = L + G$ Sum Law

$\hookrightarrow \lim_{n \rightarrow \infty} (a_n - b_n) = L - G$ Difference Law

$\hookrightarrow \lim_{n \rightarrow \infty} c a_n = c L$ Constant Law

$\hookrightarrow \lim_{n \rightarrow \infty} (a_n b_n) = L G$ Product Law

$\hookrightarrow \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{L}{G}$, $G \neq 0$ Quotient Law

$\hookrightarrow \lim_{n \rightarrow \infty} a_n^p = L^p$ for $a_n > 0$, $p > 0$ Power Law

$\hookrightarrow$ Squeeze Theorem:

If $a_n \leq b_n \leq c_n$ for $n \geq n_0$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$
then $\lim_{n \rightarrow \infty} b_n = L$

$\hookrightarrow$ If $\lim_{n \rightarrow \infty} |a_n| = 0$,
then $\lim_{n \rightarrow \infty} a_n = 0$.

↳ If $\lim_{n \rightarrow \infty} a_n = L$ and f is cont. at L
then $\lim_{n \rightarrow \infty} f(a_n) = f(L)$

↳ The sequence $\{r^n\}$ is convergent if $-1 < r < 1$ and divergent for all other values of r , i.e.

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} 0 & \text{if } -1 < r < 1 \\ 1 & \text{if } r = 1 \text{ (not } -1) \end{cases}$$

↳ Monotonic Sequence Theorem:

If a sequence is monotonic, bounded,
then it converges

↳ Particularly, increasing + bounded above $\rightarrow$ Converge
decreasing + bounded below $\rightarrow$ Converge

◦ Important Limits

$$\hookrightarrow \lim_{n \rightarrow \infty} x^n = 1, \text{ for } x > 0$$

$$\hookrightarrow \lim_{n \rightarrow \infty} x^n = 0, \text{ if } |x| < 1$$

$$\hookrightarrow \lim_{n \rightarrow \infty} \frac{1}{n^a} = 0, \text{ for } a > 0$$

$$\hookrightarrow \lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0, \text{ } x \text{ is a const.}$$

$$\hookrightarrow \lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$$

$$\hookrightarrow \lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$$

$$\hookrightarrow \lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$$

$$\hookrightarrow \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x, \text{ } x \text{ is a const.}$$

◦ Recursive sequences

$$\hookrightarrow \text{If } \lim_{n \rightarrow \infty} a_n = L, \lim_{n \rightarrow \infty} a_{n+1} = L$$

$$\text{Ex. } a_n = \sqrt{6 + a_{n-1}}$$

11.2 Series

Series

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \dots,$$

The n th partial sum is: $s_n = \sum_{i=1}^n a_i = a_1 + a_2 + \dots + a_n$

If $\{s_n\}$ is convergent and $\lim_{n \rightarrow \infty} s_n = s$ exists,

$\sum a_n$ is convergent, $a_1 + a_2 + \dots + a_n + \dots = s$ or $\sum_{n=1}^{\infty} a_n = s$

Otherwise if $\{s_n\}$ is divergent, then the series is divergent.

Geometric Series:

$$\hookrightarrow a + ar + ar^2 + ar^3 + \dots + ar^{n-1} + \dots = \sum_{n=1}^{\infty} ar^{n-1}, a \neq 0$$

$$\hookrightarrow \text{Partial sum: } s_n = \frac{a(1-r^n)}{1-r}$$

$\hookrightarrow$ If $|r| < 1$, the sum is convergent,

$$s = \frac{a}{1-r}, \text{ otherwise } (|r| \geq 1) \text{ the sum is divergent.}$$

Harmonic Series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots \quad \boxed{\text{diverges}}$$

Properties / Theorems

$\hookrightarrow$ If $\sum_{n=1}^{\infty} a_n$ is convergent,

$\hookrightarrow \lim_{n \rightarrow \infty} a_n = 0$ does not mean that $\sum_{n=1}^{\infty} a_n$ diverges

$$\text{Then } \lim_{n \rightarrow \infty} a_n = 0$$

$\hookrightarrow$ If $\lim_{n \rightarrow \infty} a_n$ DNE or $\neq 0$,

Then $\sum_{n=1}^{\infty} a_n$ diverges

$$\hookrightarrow \sum_{n=1}^{\infty} ca_n = c \sum_{n=1}^{\infty} a_n$$

$$\hookrightarrow \sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

11.3 Integral test and estimating sums

Integral Test

Let f is continuous, decreasing and positive on $[1, \infty)$, and $a_n = f(n)$

Then if $\int_1^{\infty} f(x) dx$ converges $\rightarrow \sum_{n=1}^{\infty} a_n$ converges

if $\int_1^{\infty} f(x) dx$ diverges $\rightarrow \sum_{n=1}^{\infty} a_n$ diverges.

p-series

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$, diverges if $p \leq 1$

Remainder estimate for integral

Let $f(k) = a_k$, if f is cont, +ve, dec. for $x \geq n$ and $\sum a_n$ converges

$$R_n = s - s_n$$

Then $\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx$

$$\text{and } s_n + \int_{n+1}^{\infty} f(x) dx \leq s \leq s_n + \int_n^{\infty} f(x) dx$$

11.4 Comparison Tests

Direct comparison test

Suppose $\sum a_n$ and $\sum b_n$ are series with +ve terms

Let $\sum b_n$ is convergent and $a_n \leq b_n$ for all $n \rightarrow \sum a_n$ converges

Let $\sum b_n$ is divergent and $a_n \geq b_n$ for all $n \rightarrow \sum a_n$ diverges

Limit comparison test

Suppose $\sum a_n$ and $\sum b_n$ are positive series,

1.) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$, $c > 0 \rightarrow \sum a_n$ and $\sum b_n$ both diverge or converge

2.) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$, if $\sum b_n$ converges $\rightarrow \sum a_n$ converges

3.) If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$, if $\sum b_n$ diverges $\rightarrow \sum a_n$ diverges

◦ Estimating Sums for direct comparison test

Suppose $a_n \leq b_n$ for all n ,

R_n is the remainder for $\sum a_n = s - s_n$

T_n is the remainder for $\sum b_n = t - t_n$

Then $R_n \leq T_n$ for all n .

◦ 11.5 Alternating series and absolute convergence

◦ Alternating series test

↳ If $\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \dots$, $b_n > 0$

and 1. $b_{n+1} \leq b_n$ for all n (decreasing)

2. $\lim_{n \rightarrow \infty} b_n = 0$

Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is convergent.

◦ Alternating Series estimation theorem.

if $s = \sum (-1)^{n-1} b_n$, $b_n > 0$ and satisfies

1. $b_{n+1} \leq b_n$ and 2. $\lim_{n \rightarrow \infty} b_n = 0$

Then $|R_n| = |s - s_n| \leq b_{n+1}$

◦ Absolute convergence / Conditional convergence

↳ Abs. convergence: If $\sum |a_n|$ converges, $\sum a_n$ is absolute convergent (also convergent)

Conditional convergence: If $\sum a_n$ converges and $\sum |a_n|$ diverges, $\sum a_n$ is conditionally convergent

↳ If $\sum a_n$ is conditionally convergent rearranging terms give any real numbers

◦ 11.6 Ratio and Root Test

◦ Ratio test

1.) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1 \rightarrow \sum_{n=1}^{\infty} a_n$ is abs. convergent (also convergent)

2.) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L > 1$ or $\infty \rightarrow \sum_{n=1}^{\infty} a_n$ is divergent

3.) If $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1 \rightarrow$ Inconclusive

◦ Root Test

1.) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L < 1 \rightarrow \sum_{n=1}^{\infty} a_n$ is abs convergent
(also convergent)

2.) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L < 1$ or $\infty \rightarrow \sum_{n=1}^{\infty} a_n$ is divergent

3.) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1 \rightarrow$ inconclusive

◦ 11.8 Power Series

◦ Power Series

About 0:

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots$$

About a:

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + c_3 (x-a)^3 + \dots$$

↳ Always converges when $x=a$, $= c_0$

◦ Radius, interval of convergence

↳ Given a power series $\sum_{n=0}^{\infty} c_n (x-a)^n$, the 3 possibilities are:

1.) The series converges only for $x=a$

2.) The series converges for all x

3.) The series converges for $|x-a| < R$, R is a real number

↳ R is the radius of convergence,

↳ $^a[a-R, a+R]$ is the interval of convergence

Always check end points, interval can be determined via root/ratio tests.

◦ 11.9 Representation of Functions of Power series

If $f(x)$ has $R > 0$,

Then, $f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots = \sum_{n=0}^{\infty} c_n (x-a)^n$ on interval $(a-R, a+R)$

and i.) $f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \dots = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}$

ii.) $\int f(x) dx = C + c_0(x-a) + c_1 \frac{(x-a)^2}{2} + c_2 \frac{(x-a)^3}{3} + \dots$

$$= C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

both with same radius of convergence R .

11.10 Taylor and Maclaurin Series

Taylor and Maclaurin Series



$$\begin{aligned} \text{(Taylor) Series: } f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \\ &= f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots \end{aligned}$$



$$\begin{aligned} \text{(Maclaurin) Series: } f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n \\ &= f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \dots, \text{ basically Taylor series at } a=0 \end{aligned}$$

Remainder and Equality to Taylor Series

↳ If $f(x) = T_n(x) + R_n(x)$, where T_n is the n th-degree Taylor Polynomial.

If $\lim_{n \rightarrow \infty} R_n(x) = 0$ for $|x-a| < R$, then $f(x) = \text{Taylor Series on } |x-a| < R$

↳ Taylor's inequality:

Suppose $|f^{(n+1)}(x)| \leq M$ for $|x-a| \leq d$,

then $|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}$ for $|x-a| \leq d$

↳ Forms of $R_n(x)$

Taylor's term:

$$R_n(x) = \frac{1}{n!} \int_a^x (x-t)^n f^{(n+1)}(t) dt$$

Lagrange's term! (Mean value theorem for integrals)

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!} (x-a)^{n+1} \quad \text{for } z \in (x, a)$$

Common Maclaurin Series:

$$\hookrightarrow \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots, R=1$$

$$\hookrightarrow e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, R=\infty, \text{ note } e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

$$\hookrightarrow \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots, R=\infty$$

$$\hookrightarrow \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots, R=\infty$$

$$\hookrightarrow \tan^{-1} x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots, R=1$$

$$\hookrightarrow \ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, R=1$$

$\hookrightarrow$ Binomial Series (technically also 11.12)

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3 + \dots, R=1$$

where $\binom{k}{n} = \frac{k(k-1)(k-2)\dots(k-n+1)}{n!}$, the binomial coefficient

11.1 Applications of Taylor Polynomials

Estimation by Taylor and Maclaurin Series

$\hookrightarrow$ If $f(x) = \text{sum of Taylor (or Maclaurin) Series at } a:$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(a) (x-a)^n}{n!}$$

$$= T_n(x) \text{ as } n \rightarrow \infty$$

then $f(x) \approx T_n(x)$, n of your choice

Size of error:

1.) If series alternates, find $n+1$ th term, $|R_n(x)| \leq |a_{n+1}|$

2.) In every case, we can apply Taylor's inequality,

$$\text{Let } |f^{(n+1)}(x)| \leq M,$$

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1}$$

Binomial approximation

$\hookrightarrow$ Express term in form $(1+x)^k$

$$\text{ex } \frac{1}{(2+x)^3} = (2+x)^{-3} = 2^{-3} (1+\frac{x}{2})^{-3}$$

$\hookrightarrow$ Expand as binomial series

$$\text{ex } \frac{1}{(2+x)^3} = 2^{-3} \left[1 + (-3)\left(\frac{x}{2}\right) + \frac{(-3)(-4)}{2!} \left(\frac{x}{2}\right)^2 + \dots \right] \rightarrow 0$$

$\hookrightarrow$ Consider variable $\gg n$ or $\ll n$, take appropriate terms as indicated

$$\text{ex for } 0 < x \ll 1 \quad \frac{1}{(2+x)^3} \approx 2^{-3} \left[1 + \left(-\frac{3x}{2}\right) \right] = \frac{1}{8} - \frac{3x}{16}$$

11.3 Fourier Series

Fourier Series



Series:

- For piecewise, continuous, periodic functions

- For period 2π , on $[-\pi, \pi]$: $f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$, where

$$\hookrightarrow a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$\hookrightarrow$ Fourier coefficients:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

- For period $2L$, on $[-L, L]$: $f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right]$

$$\hookrightarrow a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$\hookrightarrow$ Fourier coefficients:

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

- If expressed as T , $[-\frac{T}{2}, \frac{T}{2}]$: $f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{2n\pi x}{T}\right) + b_n \sin\left(\frac{2n\pi x}{T}\right) \right]$,

$$\text{For } \frac{2\pi}{T} = \omega, \quad f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega x) + b_n \sin(n\omega x) \right]$$

$$\hookrightarrow a_0 = \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x) dx$$

$\hookrightarrow$ Fourier coefficients:

$$a_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x) \cos\left(\frac{2n\pi x}{T}\right) dx, \quad b_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x) \sin\left(\frac{2n\pi x}{T}\right) dx$$

$$\text{For } \frac{2\pi}{T} = \omega,$$

$$a_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x) \cos(n\omega x) dx, \quad b_n = \frac{2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(x) \sin(n\omega x) dx$$

Equality and convergence of Fourier Series

- $\hookrightarrow$ It: 1. f is periodic,
2. With period T ,
3. f, f' are piecewise cont. on $[-\frac{T}{2}, \frac{T}{2}]$

Then: 1. $f(x) =$ sum of Fourier series at all numbers x where f is cont.

2. At x where f is not continuous,

$$\text{Sum of Fourier series} = \frac{1}{2} [f(x^+) + f(x^-)]$$

12.4 The Cross Product

(Technically not in syllabus, but might help)

Properties of $\times$ product

1. $a \times b = -b \times a$
2. $(ca) \times b = c(a \times b) = a \times (cb)$
3. $a \times (b+c) = a \times b + a \times c$
4. $(a+b) \times c = a \times c + b \times c$
5. $a \cdot (b \times c) = (a \times b) \cdot c$
6. $a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$

12.5 Equations of lines and planes

Equations of a line

- Vector equation: $r = r_0 + tv$ or $\langle x, y, z \rangle = \langle x_0 + at, y_0 + bt, z_0 + ct \rangle$
 $r_0 = \langle x_0, y_0, z_0 \rangle$, $v = \langle a, b, c \rangle$

- Parametric equations:

$$x = x_0 + at$$

$$y = y_0 + bt$$

$$z = z_0 + ct$$

- Symmetric equations:

$$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c} \text{ for } a, b, c \neq 0$$

Line segment

↳ A line segment from r_0 to r_1 is:

$$r(t) = (1-t)r_0 + tr_1 \quad 0 \leq t \leq 1$$

Equations of plane

- Vector equation of plane:

$$n \cdot (r - r_0) = 0 \quad \text{or} \quad n \cdot r = n \cdot r_0$$

n is the normal vector

Distance b/w point, plane:

$$D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}} \text{ for point } p(x_1, y_1, z_1) \text{ and plane } ax + by + cz + d = 0$$

- Scalar equation of plane:

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

$$\text{or } ax + by + cz + d = 0, d = -(ax_0 + by_0 + cz_0)$$

Distance b/w parallel planes:

$$\text{For } p_1: ax + by + cz + d_1 = 0$$

$$p_2: ax + by + cz + d_2 = 0$$

$$D = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$$

12.6 Cylinders and Quadric Surfaces

Quadric surfaces

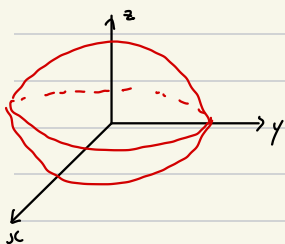
- General form, $Ax^2 + By^2 + Cz^2 + Dx + Ey + Fz + Gx + Hy + Iz + J = 0$
- Standard form: $Ax^2 + By^2 + Cz^2 + J = 0$ or $Ax^2 + Dy^2 + Iz^2 = 0$

Graphs, equations, traces of quadric surfaces

Ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Traces: all ellipses

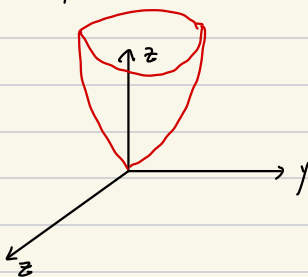


Elliptic Paraboloid:

power of 1 → axis of paraboloid

$$\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

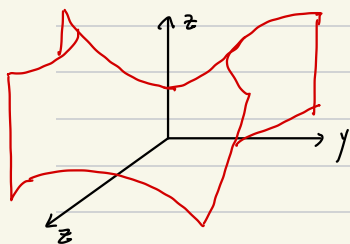
Traces: Horizontal traces → ellipses
Vertical traces → parabolas



Hyperbolic paraboloid:

$$\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

Traces: Horizontal → hyperbola

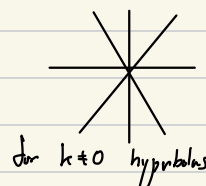
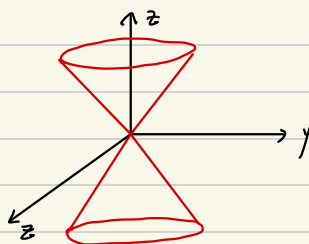


Cone:

axis of symmetry

$$\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

Traces: Horizontal → ellipse
Vertical → for $k=0$ pair of lines



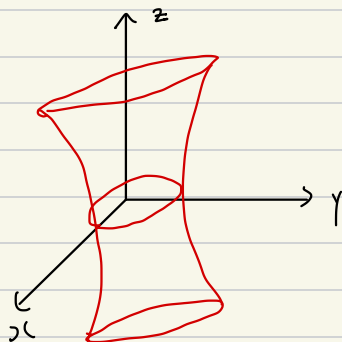
for $k \neq 0$ hyperbolas

Hyperboloid of one sheet:

negative axis of symmetry

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

Traces: Horizontal → ellipses
Vertical → hyperbolas



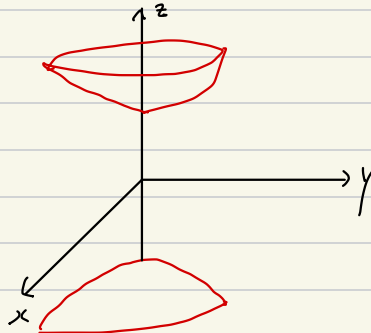
Hyperboloid of two sheets:

Two negative → Two sheets

$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

axis of symmetry

Traces: Horizontal → ellipses for $k > c/k < -c$
(for $\frac{z^2}{c^2}$ in this case)
Vertical → parabolas



13.1 Vector function and Space Curves

Limit and continuity

- If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$

then $\lim_{t \rightarrow a} \mathbf{r}(t) = \langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \rangle$

- Continuous if $\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$

Space curves

$$x = f(t) \quad y = g(t) \quad z = h(t)$$

13.2 Derivatives and Integrals of vector functions

Derivative

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} \quad \text{"first principle"}$$

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$$

Differentiation rules

If u and v are differentiable vector functions, c is a scalar, f is a ^{"scalar function"} real-valued function, then

1. $\frac{d}{dt} [u(t) + v(t)] = u'(t) + v'(t)$

2. $\frac{d}{dt} [c u(t)] = c u'(t)$

3. $\frac{d}{dt} [f(t) u(t)] = f'(t) u(t) + f(t) u'(t)$

4. $\frac{d}{dt} [u(t) \cdot v(t)] = u'(t) \cdot v(t) + u(t) \cdot v'(t)$

5. $\frac{d}{dt} [u(t) \times v(t)] = u'(t) \times v(t) + u(t) \times v'(t)$

6. $\frac{d}{dt} [u(f(t))] = f'(t) u'(f(t))$

Theorem: If $|\mathbf{r}(t)| = c$, $\mathbf{r}'(t)$ is $\perp$ to $\mathbf{r}(t)$ for all t

Integrals

$$\int_a^b \mathbf{r}(t) dt = \int_a^b f(t) dt \mathbf{i} + \int_a^b g(t) dt \mathbf{j} + \int_a^b h(t) dt \mathbf{k}$$

13.3 Arc Length and Curvature

Arc length

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt \quad \hookrightarrow s(t) = \int_a^t |\mathbf{r}'(u)| du$$
$$= \int_a^b |\mathbf{r}'(t)| dt$$

Arc length function

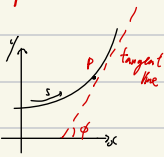
$$\hookrightarrow \frac{ds}{dt} = |\mathbf{r}'(t)|$$

Unit tangent vector

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

Curvature

Def'n 1: Geometry:

$$\kappa = \left| \frac{d\phi}{ds} \right|$$


Def'n 2: Tangent

$$\kappa = \left| \frac{dT}{ds} \right| = \frac{|T'(t)|}{|r'(t)|}$$

Def'n 3: Cross product

$$\kappa = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}$$

Def'n 4: $y = f(x)$

$$\kappa = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}$$

Def'n 5: $x = x(t)$
 $y = y(t)$

$$\kappa = \frac{|y''x' - y'x''|}{[x'^2 + y'^2]^{3/2}}$$

Normal, binormal vector

$$\text{Normal: } N(t) = \frac{T'(t)}{|T'(t)|}$$

$$\text{Binormal: } B(t) = T(t) \times N(t)$$

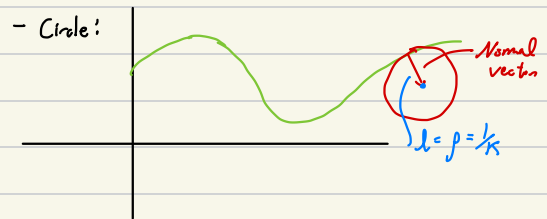
Normal, osculating plane

Normal: Defined by N, B vectors
 $\perp$ to T

Osculating: Defined by N, T vectors
 $\perp$ to B

Circle, radius of curvature

$$\text{radius } \rho = \frac{1}{\kappa}$$



13.4 Velocity and Acceleration

Projectile Motion

$$x = (v_0 \cos \alpha) t$$

$$y = (v_0 \sin \alpha) t - \frac{1}{2} g t^2$$

Tangential and Normal Components of Acceleration

Let v be magnitude of $v(t) = |r'(t)|$

$$\text{Tangential: } a_T = r'(T)$$

$$\text{Normal: } a_N = \kappa v^2(N)$$

$$\text{Acceleration: } a = a_T + a_N = r'(T) + \kappa v^2(N)$$

14.1 Function of Several Variables

Level curves

A level curve of f is a curve of eq. $f(x, y) = k$, k is a constant.

14.2 Limits and Continuity

Precise Definition of 2D limit

$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$ implies for every $\epsilon > 0$ there exists a $\delta > 0$ s.t.

$$\text{if } \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x,y) - L| < \epsilon$$

Limit DNE

If $f(x,y) \rightarrow L_1$ as $(x,y) \rightarrow (a,b)$ along path C_1 , $L_1 \neq L_2$ then $\lim_{(x,y) \rightarrow (a,b)}$ DNE
 $f(x,y) \rightarrow L_2$ as $(x,y) \rightarrow (a,b)$ along path C_2

- Continuity

If $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$, then f is cont @ (a,b)

if for all $(a,b) \in D$ f is cont, f is cont at every point in D .

- Higher dimensional limits

$\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = L$ if for all $\epsilon > 0$, there exists a $\delta > 0$ s.t.

$$|\vec{x} - \vec{a}| < \delta \Rightarrow |f(\vec{x}) - L| < \epsilon$$

- 14.3 Partial derivatives

- Partial Derivatives

$$f_x(x,y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x,y)}{h}, \text{ wrt. } x$$

$$f_y(x,y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x,y)}{h}, \text{ wrt. } y$$

↳ Notation: $f_x = \frac{\partial f}{\partial x}$
 $f_y = \frac{\partial f}{\partial y}$

- Clairaut's Theorem

If f is defined on disk D w/ point (a,b) , if f_{xy} and f_{yx} are both cont. on D , then:

$$f_{xy}(a,b) = f_{yx}(a,b)$$

and $f_{xxy}(a,b) = f_{yyx}(a,b) = f_{xyx}(a,b)$ etc.

- 14.4 Tangent Planes and Linear Approximations

- Tangent plane to curve

For $z = f(x,y)$, @ point $P(x_0, y_0, z_0)$, tangent plane is:

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

- Linear approximations and differentials

Differentials for $z = f(x,y)$

$$dz = f_x(x,y)dx + f_y(x,y)dy = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy, \text{ } dx = \Delta x, dy = \Delta y$$

For $\Delta x = x - a, \Delta y = y - b$

$$dz = f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

Linear approximation at (a,b) is:

$$f(x,y) \approx f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) \\ \approx f(a,b) + dz$$

• Differentiability of $f(x,y)$

$f(x,y)$ is diff for $\vec{x} = (x,y)$, $h = h_1, h_2$, if
 $f(\vec{x} + \vec{h}) - f(\vec{x}) = \nabla f(\vec{x}) \cdot \vec{h} + o(|\vec{h}|)$, where $\nabla f(\vec{x})$ is the **gradient**.

If $\frac{|o(|\vec{h}|)|}{|\vec{h}|} \rightarrow 0$ as $\vec{h} \rightarrow \vec{0}$, typically, rewrite $o(h)$ as dot product to conclude by squeeze theorem.

Alternatively, if f_x, f_y exists near (a,b) and $f(a,b)$ are cont.
 then f is diff at (a,b)

• 14.5 The Chain Rule

• Case 1: $x = g(t)$, $y = h(t)$, $z = f(x,y)$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

• Case 2: $z = f(x,y)$, $x = g(s)$, $y = h(s)$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$

• General chain rule:

For $z = f(x,y)$, if u is a diff function of n variables: $x_1, x_2, x_3 \dots x_n$ and x_j is a diff function of m variables $t_1, t_2 \dots t_m$, then:

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \cdot \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \cdot \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \cdot \frac{\partial x_n}{\partial t_i}$$

• Implicit differentiation

↳ for $F(x,y) = 0$, $y = h(x)$

$$\frac{dy}{dx} = - \frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = - \frac{F_x}{F_y}$$

↳ for $F(x,y,z) = 0$, $z = f(x,y)$

$$\frac{\partial z}{\partial x} = - \frac{F_x}{F_z}, \quad \frac{\partial z}{\partial y} = - \frac{F_y}{F_z}$$

• 14.6 Directional Derivatives and Gradient Vector

• Gradient Vector

$$\nabla f(x,y), \text{ Gradient} = \langle f_x, f_y \rangle$$

• Directional derivative

The directional derivative of f at (x_0, y_0) in direction of unit vector $u = \langle a, b \rangle$ is:

$$D_u f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ah, y_0 + bh) - f(x_0, y_0)}{h}$$

$$= f_x(x_0, y_0)a + f_y(x_0, y_0)b$$

$$= \nabla f(x_0, y_0) \cdot u = \nabla f(x_0, y_0) \cdot \langle a, b \rangle$$

• Tangent and Normal to level curve

↳ For $f(x, y) = 0$,

$$\nabla f(\vec{r}) \cdot \vec{r} = 0 \rightarrow \text{Gradient} \perp \text{Tangent}$$

↳ Tangent:

$$(x - x_0, y - y_0) \cdot \nabla f = 0$$

$$(x - x_0) f_x(x_0, y_0) + (y - y_0) f_y(x_0, y_0) = 0$$

↳ Normal, if $\vec{r} = \left(\frac{\partial f}{\partial y}, -\frac{\partial f}{\partial x} \right)$:

$$(x - x_0, y - y_0) \cdot \vec{r} = 0$$

$$(x - x_0) f_y(x_0, y_0) - (y - y_0) f_x(x_0, y_0) = 0$$

• Tangent plane and normal line to level surface

For $F(x, y, z) = k$, a level surface for $F(x, y, z)$

↳ Tangent plane @ $P(x_0, y_0, z_0)$

$$\nabla F(\vec{r}_0) \cdot \vec{r}(\vec{r}_0) = 0$$

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

↳ Normal line:

$$x, y, z = (x_0, y_0, z_0) + t(F_x, F_y, F_z)$$

$$\frac{x - x_0}{F_x(x_0, y_0, z_0)} = \frac{y - y_0}{F_y(x_0, y_0, z_0)} = \frac{z - z_0}{F_z(x_0, y_0, z_0)}$$

• Properties of Gradient

↳ ∇f points in the direction of maximum rate of increase of f at $\vec{r}$,

such maximum rate of change is $|\nabla f(\vec{r})|$

↳ $\nabla f \perp$ to the level curve or level surface through $\vec{r}$

• 14.7 Maximum and Minimum Values

• Local maximum / Local minimum

↳ Either $f_x(a, b) = 0$ and $f_y(a, b) = 0$

or $f_x(a, b)$ DNE and $f_y(a, b)$ DNE

(a, b) is known as the critical point

• Second derivative test

$$\text{Determinant } D = f_{xx}f_{yy} - (f_{xy})^2$$

↳ If $D < 0$, (a, b) is a saddle point

$D > 0$, $f_{xx} > 0$, (a, b) is a min

$D < 0$, $f_{xx} < 0$, (a, b) is a max

$D = 0 \rightarrow$ inconclusive

◦ Absolute Maximum and Minimum

↳ How to find:

1. Find f at critical points of f
2. Find the extreme values of f on boundary D
3. Find largest, smallest

◦ 14.8 Lagrange Multipliers

◦ One constraint

For $f(x, y, z)$ subject to $g(x, y, z) = k$, k is a constant

Find x, y, z and λ s.t.

$$\textcircled{1}: f_x = \lambda g_x$$

$$\textcircled{2}: f_y = \lambda g_y$$

$$\textcircled{3}: f_z = \lambda g_z$$

$$\textcircled{4}: g(x, y, z) = k$$

◦ Two constraints

For $f(x, y, z)$ subject to $g(x, y, z) = k$, $h(x, y, z) = l$, k, l is a constant

Find x, y, z and λ, μ s.t.

$$\textcircled{1}: f_x = \lambda g_x + \mu h_x$$

$$\textcircled{2}: f_y = \lambda g_y + \mu h_y$$

$$\textcircled{3}: f_z = \lambda g_z + \mu h_z$$

$$\textcircled{4}: g(x, y, z) = k$$

$$\textcircled{5}: h(x, y, z) = l$$

◦ (14.9?) Reconstructing a Function from its Gradient

◦ Gradient?

Suppose we have $P(x, y)$ and $Q(x, y)$

$[P(x, y), Q(x, y)]$ is a gradient iff

$$\frac{\partial P(x, y)}{\partial y} = \frac{\partial Q(x, y)}{\partial x}$$

◦ Reconstruction

$$\hookrightarrow \text{If } G(x, y) = f_x, \text{ then } \int G(x, y) dx + \phi(y) = f(x, y)$$

$$\hookrightarrow \text{If } H(x, y, z) = f_x, \text{ then } \int H(x, y, z) dx + \phi(y) + \lambda(z) = f(x, y, z)$$

◦

$A(x, y, z), B(x, y, z), C(x, y, z)$,

$[A, B, C]$ is a gradient iff

$$\frac{\partial A}{\partial y} = \frac{\partial B}{\partial x}, \quad \frac{\partial A}{\partial z} = \frac{\partial C}{\partial x}$$

$$\frac{\partial B}{\partial x} = \frac{\partial A}{\partial y}, \quad \frac{\partial B}{\partial z} = \frac{\partial C}{\partial y}$$

$$\frac{\partial C}{\partial x} = \frac{\partial A}{\partial z}, \quad \frac{\partial C}{\partial y} = \frac{\partial B}{\partial z}$$

14.10 Differentiability of an Integral wrt. its Parameter

For $F(x) = \int_c^d f(x, y) dy$

Here's a cool trick:

for $x \in [a, b]$ and $y \in [c, d]$, and $f(x, y)$ has a cont. derivative wrt x , for $x \in [a, b]$

$$\frac{dF}{dx} = \frac{d}{dx} \int_c^d f(x, y) dy = \int_c^d \frac{\partial f}{\partial x} dy$$

Leibnitz's Rule

↳ In region R of xy plane, for which $\phi_1(x)$ and $\phi_2(x)$ is cont., diff wrt x and $f(x, y)$ is cont., diff

$$\text{Let } F(x) = \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) dy$$

$$\text{then: } \frac{dF}{dx} = \int_{\phi_1(x)}^{\phi_2(x)} \frac{\partial f}{\partial x} dy + f(x, \phi_2(x)) \frac{d\phi_2}{dx} - f(x, \phi_1(x)) \frac{d\phi_1}{dx}$$